

- 8 (a) splitting of a heavy nucleus (*not atom/nuclide*)
into two (lighter) nuclei of approximately same mass M1
A1 [2]
- (b) ${}^1_0\text{n}$
 ${}^4_2\text{He}$ (*allow* ${}^4_2\alpha$) M2
 ${}^7_3\text{Li}$ A1 [3]
- (c) emitted particles have kinetic energy B1
range of particles in the control rods is short / particles stopped in rods /
lose kinetic energy in rods B1
kinetic energy of particles converted to thermal energy B1 [3]

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Section B

9 (c) (i) non-inverting (amplifier)

B1 [1]